

Mathematical Derivation of the TwoTimescaleResilience Framework: Species-Aware Pipeline with Exposure and Mode of Action Mapping

Model Documentation

May 27, 2026

Abstract

This document provides a full mathematical derivation of the `TwoTimescaleResilience.jl` spatial framework. It explicitly details how raw environmental rasters are translated into species-specific vulnerabilities. By introducing a target group index (q), we trace the pipeline from raw ambient stress, through species-specific exposure filters, intermediate Modes of Action (MoA), and finally to Dynamic Energy Budget (DEB) physiological axes. The derivation concludes with the core integration theorem, proving that amplification of a perturbation response is inversely proportional to the background-conditioned restoring force. The narrative is supported by a synthetic 3×3 spatial grid example.

1 Phase 1: Species-Aware Exposure Filtering

The framework operates over a discrete spatial grid where each cell (g, m) represents a location and time. Let the raw ambient environmental conditions be represented by a vector of variables $\mathbf{R}_{g,m}$. For our example, we use a pathogen proxy and an organic (BOD) proxy:

$$\mathbf{R}_{g,m} = \begin{bmatrix} R_{\text{pathogen}} \\ R_{\text{organic}} \end{bmatrix}_{g,m} \quad (1)$$

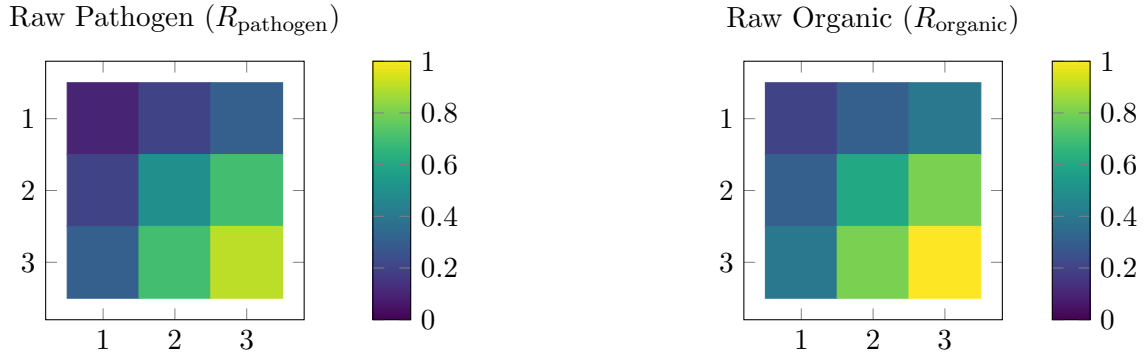


Figure 1: Input raw environmental stress fields.

A central feature of the framework is its support for multiple species or target groups (e.g., humans, avian species, aquatic organisms). We denote a specific target group with the superscript (q). Because organisms interact with their environment differently, raw variables are filtered into *effective exposure* $e^{(q)}$ via contact, habit, or use multipliers $h_j^{(q)}$:

$$e_{j,g,m}^{(q)} = h_j^{(q)} R_{j,g,m} \quad (2)$$

For a fully aquatic species, contact is continuous ($h_j \approx 1.0$). For humans, contact is filtered by drinking water infrastructure, sanitation, and behavioral avoidance ($h_j < 1.0$).

2 Phase 2: Mode of Action to DEB Axes Mapping

Rather than mapping raw exposure directly to organism health, the pipeline processes effective exposure through two intermediate biological layers.

First, the effective exposure vector is mapped to intermediate **Modes of Action** $\mathbf{m}^{(q)}$ (e.g., thermal stress, oxygen depletion, direct toxicity, immune load). This mapping utilizes a linear weight matrix $U^{(q)}$ and a pairwise interaction matrix $\Gamma^{(r,q)}$:

$$m_{r,g,m}^{(q)} = \sum_j U_{rj}^{(q)} e_{j,g,m}^{(q)} + \sum_{j < k} \Gamma_{jk}^{(r,q)} e_{j,g,m}^{(q)} e_{k,g,m}^{(q)} \quad (3)$$

Next, these MoA values are mapped to the four core **DEB physiological axes** $\mathbf{s}^{(q)}$: Assimilation (s_A), Maintenance (s_M), Growth (s_G), and Reproduction (s_R). This step uses a secondary weight matrix $W^{(q)}$ and interactions $\tilde{\Gamma}^{(a,q)}$:

$$s_{a,g,m}^{(q)} = \sum_r W_{ar}^{(q)} m_{r,g,m}^{(q)} + \sum_{r < k} \tilde{\Gamma}_{rk}^{(a,q)} m_{r,g,m}^{(q)} m_{k,g,m}^{(q)} \quad (4)$$

By separating MoA from DEB axes, the model easily adapts to different taxa. For instance, hypoxia (an MoA) might heavily penalize assimilation (s_A) in fish, but have zero effect on a terrestrial bird. The resulting matrices represent the target group's physiological cost field.

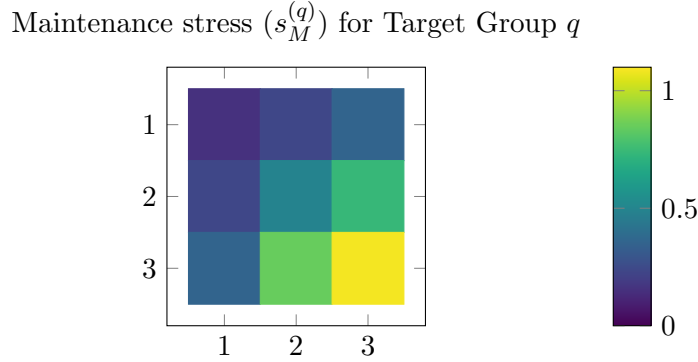


Figure 2: The calculated physiological maintenance cost raster for species (q), accounting for species-specific exposure filters and MoA mappings.

3 Phase 3: Adaptive Margin and Restoring Force

The aggregate physiological burden depletes the baseline adaptive capacity ($A_0^{(q)}$) of the species. The scalar **Adaptive Margin** (A) incorporates the DEB axes via sensitivity weights $\alpha^{(q)}$. An optional dynamic condition/reserve buffer $Z_{g,m}^{(q)}$ can also be integrated to account for historical depletion or buffering:

$$A_{g,m}^{(q)} = A_0^{(q)} + \omega_Z^{(q)} Z_{g,m}^{(q)} - \sum_{a \in \{A, M, G, R\}} \alpha_a^{(q)} s_{a,g,m}^{(q)} \quad (5)$$

The adaptive margin directly determines the **Restoring Force** ($\lambda^{(q)}$), representing the organism's exponential recovery rate following a perturbation. This is modeled via a bounded Michaelis-Menten formulation:

$$\lambda_{g,m}^{(q)} = \lambda_{\min}^{(q)} + \left(\lambda_{\max}^{(q)} - \lambda_{\min}^{(q)} \right) \frac{[A_{g,m}^{(q)}]_+}{K_A^{(q)} + [A_{g,m}^{(q)}]_+}, \quad \text{where } [A]_+ = \max(A, 0) \quad (6)$$

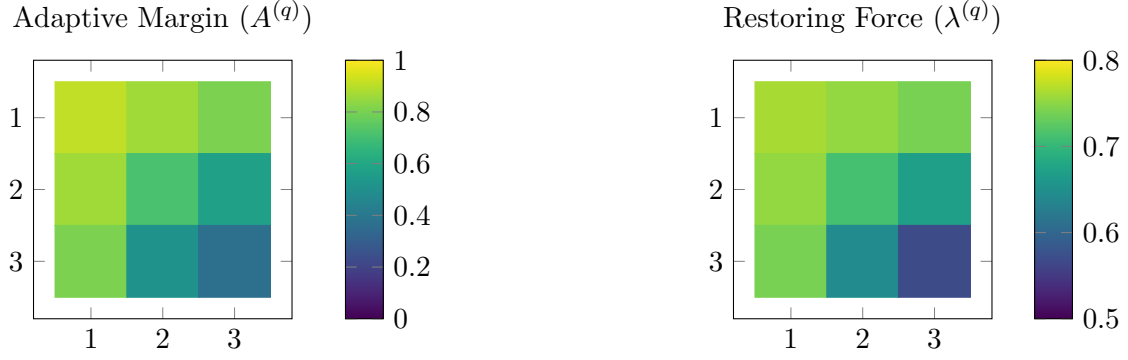


Figure 3: Higher physiological costs deplete the adaptive margin, thereby weakening the target group's restoring force.

4 Phase 4: Deriving the Amplification Factor

Reduced restoring force amplifies the impact of short-term acute perturbations. Let an identical standardized event impose a physiological cost $C_{\text{event}}(t)$. The organism's displacement $y(t)$ from homeostasis is governed by:

$$\frac{dy}{dt} = -\lambda_{g,m}^{(q)} y + q_c C_{\text{event}}(t), \quad y(0) = 0 \quad (7)$$

The analytical solution for the displacement trajectory is:

$$y_{g,m}(t) = q_c \int_0^t e^{-\lambda_{g,m}^{(q)}(t-u)} C_{\text{event}}(u) du \quad (8)$$

We define the total **Response Burden** as the Area Under the Curve (AUC) of the displacement:

$$\text{AUC}_y(g, m) = \int_0^\infty y_{g,m}(t) dt = q_c \int_0^\infty \left[\int_0^t e^{-\lambda_{g,m}^{(q)}(t-u)} C_{\text{event}}(u) du \right] dt \quad (9)$$

Applying Fubini's Theorem to reverse the order of integration ($0 \leq u \leq t < \infty \implies u \leq t < \infty$ and $0 \leq u < \infty$):

$$\text{AUC}_y(g, m) = q_c \int_0^\infty C_{\text{event}}(u) \left[\int_u^\infty e^{-\lambda_{g,m}^{(q)}(t-u)} dt \right] du \quad (10)$$

Evaluating the inner integral by substituting $r = t - u$:

$$\int_u^\infty e^{-\lambda_{g,m}^{(q)}(t-u)} dt = \int_0^\infty e^{-\lambda_{g,m}^{(q)} r} dr = \frac{1}{\lambda_{g,m}^{(q)}} \quad (11)$$

Thus, the total response burden simplifies to:

$$\text{AUC}_y(g, m) = \frac{q_c}{\lambda_{g,m}^{(q)}} \int_0^\infty C_{\text{event}}(u) du = \frac{q_c}{\lambda_{g,m}^{(q)}} \text{AUC}_C \quad (12)$$

The integral of the event cost (AUC_C) is identical regardless of the background state. To standardize vulnerability, we compute the ratio of the response burden in a stressed grid cell to the response burden under a pristine baseline condition ($A = A_0^{(q)} \implies \lambda_0^{(q)}$):

$$\frac{\text{AUC}_y(g, m)}{\text{AUC}_y(0)} = \frac{\frac{q_c}{\lambda_{g,m}^{(q)}} \text{AUC}_C}{\frac{q_c}{\lambda_0^{(q)}} \text{AUC}_C} = \frac{\lambda_0^{(q)}}{\lambda_{g,m}^{(q)}} \quad (13)$$

This confirms that the **Amplification Factor** ($\mathcal{F}^{(q)}$) is strictly defined by the ratio of the restoring forces:

$$\mathcal{F}_{g,m}^{(q)} = \frac{\lambda^{(q)}(A_0)}{\lambda^{(q)}(A_{\text{DEB},g,m})} \quad (14)$$

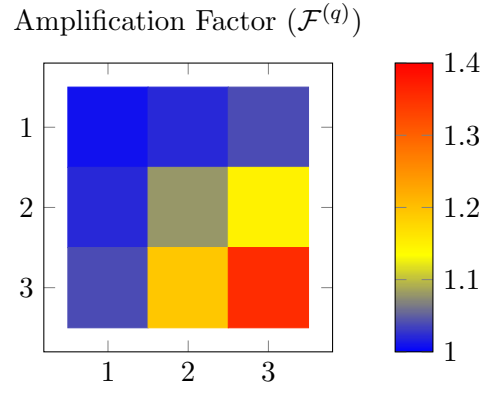


Figure 4: The final spatial result. Background stress conditions local vulnerability, magnifying the impact of future perturbations.